

Skills Summary

Solving Linear Equations with Variables on Both Sides

Use this reference while completing your worksheet or when studying.

Skill 1 — From a Balance or Table Model to an Equation

The model: a balanced pair of pans is an equation. A sealed bag $\square$ holds x counters; a loose counter $\bullet$ is 1. Whatever is on the left is worth the same as whatever is on the right.

The legal moves: remove the same number of bags from both pans (subtract kx from both sides), remove the same number of counters from both pans (subtract a constant), or split both pans into the same number of equal groups (divide). A table works the same way: two cost columns set equal to each other is a break-even equation.

Example: Left pan: 2 bags and 3 counters. Right pan: 1 bag and 7 counters.

Step 1. Both pans balance, so the two amounts are equal — that is what turns the picture into $2x + 3 = x + 7$ rather than into two separate expressions.

Step 2. Take 1 bag off each pan: $x + 3 = 7$. Take 3 counters off each pan: $x = 4$, so each bag holds 4 counters.

Try it: Left pan: 3 bags and 4 counters. Right pan: 1 bag and 10 counters. Write the equation and solve it.

$x = 4$:check

Skill 2 — Collecting the Variable Terms on One Side

Rule: an equation stays true when the same quantity is added to, or subtracted from, both sides. Use that to get every x on one side and every plain number on the other, then divide by the coefficient.

Which side? Subtract the *smaller* x -coefficient, so the variable term that survives is positive — fewer sign errors. From $7x - 2 = 4x + 13$, subtract $4x$, not $7x$.

Always check by putting the answer back into the original equation and evaluating both sides separately. They must give the same number.

Example: Solve $7x - 2 = 4x + 13$.

Step 1. Both sides carry an x term, so nothing can be isolated yet; the first move is to collect the variable on the side with the larger coefficient.

Step 2. Subtract $4x$ from both sides: $3x - 2 = 13$. Add 2 to both sides: $3x = 15$. Divide by 3 to get $x = 5$, and check: $7(5) - 2 = 33 = 4(5) + 13$.

Try it: Solve $9x + 4 = 5x - 8$.

$x = -3$:check

Skill 3 — Distribute First, Then Collect

Rule: when a side is a product such as $5(x-2)$, multiply it out before collecting — $5(x-2) = 5x - 10$. Multiply the outside factor by *every* term inside, signs included.

When the variable disappears: after collecting, an x -free sentence is left. If it is true, such as $15 = 15$, every number works and the equation has infinitely many solutions. If it is false, such as $8 = 5$, no number works and the equation has no solution.

Example: Solve $5(x-2) = 3x+4$.

Step 1. The left side is a product, so it cannot be combined with anything until it is distributed — that is the only move available first.

Step 2. Distribute: $5x - 10 = 3x + 4$. Subtract $3x$: $2x - 10 = 4$. Add 10: $2x = 14$, so $x = 7$, and $5(7-2) = 25 = 3(7) + 4$.

Try it: Solve $2(3x+1) = 4x+12$.

check: $5 = x$

(!) Watch out: a move must hit *both* sides — subtracting $2x$ from the left only is the most common error on these. And when you distribute a negative factor, the sign of every term inside changes: $-3(x-4) = -3x+12$, not $-3x-12$.